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Automated compositing of content compilations

Abstract

A system includes a computing platform having processing hardware and a memory storing a software code. The processing hardware is configured to execute the software code to receive multiple content units each including a start descriptor for an initial content segment and an end descriptor for a last content segment, identify the start descriptor and the end descriptor for each of the content units, and select a first content unit for beginning a content compilation. The processing hardware is further configured to execute the software code to determine multiple similarity metrics each comparing the end descriptor of the first content unit with the start descriptor of a respective one of the other content units, rank, using the similarity metrics, the other content units with respect to one another, select, based on the rank, a second content unit, and composite the content compilation using the second content unit.

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Background/Summary

BACKGROUND

(1) Joining or merging discrete units of content is a common task in content creation. For example, multiple video clips may be merged by a professional video editor in a studio, or by an amateur user via an online social media platform or utilizing a commercial software application. However, conventional solutions for merging discrete content units to form a content compilation achieve good results only when working with content that has a common theme or context running through the content units. For instance, where the content units are video clips, the clips being merged are typically from sporting events, celebrations, vacations, home videos, or educational tutorials, to name a few examples. In the conventional art, video clips selected from multiple different topics or themes tend to produce poor composite videos that may be jarring to watch and listen to due to abrupt transitions of visual style, thematic subject matter, and sound effects. Consequently, there is a need in the art for an automated solution for compositing disparate units of content from a broad range of topics, many of which may have different audio-visual characteristics or styles, to produce a compilation of that content that is appealing, especially where these compilations are unique for each user and human involvement cannot be afforded.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a diagram of a system for performing automated compositing of content compilations, according to one exemplary implementation;
- (2) FIG. 2 shows a flowchart presenting an exemplary method for use by a system to perform automated compositing of content compilations, according to one implementation;
- (3) FIG. 3A shows a diagram depicting extraction of an exemplary feature set from a content unit, according to one implementation;
- (4) FIG. 3B shows a diagram depicting generation of an exemplary start descriptor and an exemplary end descriptor of a content unit, according to one implementation;
- (5) FIG. 4A shows a portion of an exemplary multi-dimensional feature space including feature space landmarks, according to one implementation;
- (6) FIG. 4B shows a diagram of an exemplary content compilation trajectory among feature space landmarks, according to one implementation;
- (7) FIG. 5A shows a diagram depicting compositing of a content compilation, according to one exemplary implementation;
- (8) FIG. 5B shows a diagram depicting compositing of a content compilation, according to another exemplary implementation;
- (9) FIG. 6A shows a diagram depicting compositing of a content compilation, according to yet another exemplary implementation;
- (10) FIG. 6B shows a diagram depicting production of an implicit signal for improving content sequencing performance for compositing of a content compilation, based on usage data; and
- (11) FIG. 7 shows a flowchart presenting an exemplary method for use by a system to perform automated compositing of content compilations, according to another implementation.

DETAILED DESCRIPTION

(12) The following description contains specific information pertaining to implementations in the present disclosure. One skilled in the art will recognize that the present disclosure may be implemented in a manner different from that specifically discussed herein. The drawings in the present application and their accompanying detailed description are directed to merely exemplary implementations. Unless noted otherwise, like or corresponding elements among the figures may be indicated by like or corresponding reference numerals.

(13) The present application is directed to automated systems and methods for compositing content compilations that address and overcome the deficiencies in the conventional art. According to the present automated compositing solution, information about features of content units, such as their audio features, the images they include, and their semantic features, for example, are extracted and used to automate the compositing of the content units in a manner that produces results comparable in quality to those produced by an expert human content editor. In contrast to conventional techniques for creating content compilations, the present solution can advantageously work with content that spans a wide range of themes and topics, as well as a host of visual styles and audio tracks created by different artists in different settings and for different purposes. In addition, according to the present compositing solution a user supplied template or layout, which is typically relied upon in conventional content compilation techniques, is neither sought nor utilized. As a result, the present solution advantageously enables the automated production of a coherent content compilation without abrupt switches in visual style, story arc, or audio effects.

(14) As defined in the present application, the terms “automation,” “automated,” and “automating” refer to systems and processes that do not require human intervention. Although in some implementations a human editor may review the content compilations composited by the systems and using the methods described herein, that human involvement is optional. Thus, the methods

described in the present application may be performed under the control of hardware processing components of the disclosed automated systems.

(15) Moreover, as defined in the present application, the expression “machine learning model” may refer to a mathematical model for making future predictions based on patterns learned from samples of data or “training data.” Various learning algorithms can be used to map correlations between input data and output data. These correlations form the mathematical model that can be used to make future predictions on new input data. Such a predictive model may include one or more logistic regression models, Bayesian models, or neural networks (NNs).

(16) A “deep neural network,” in the context of deep learning, may refer to an NN that utilizes multiple hidden layers between input and output layers, which may allow for learning based on features not explicitly defined in raw data. As used in the present application, a feature identified as an NN refers to a deep neural network. In various implementations, NNs may be trained as classifiers and may be utilized to perform image processing or natural-language processing.

(17) FIG. 1 shows a diagram of system **100** for performing automated compositing of content compilations, according to one exemplary implementation. As shown in FIG. 1, system **100** includes computing platform **102** having processing hardware **104** and system memory **106** implemented as a computer-readable non-transitory storage device. According to the present exemplary implementation, system memory **106** stores software code **110**, one or more machine learning models **120** (hereinafter “machine learning model(s) **120**”), and content and feature set database **112** storing features extracted from a curated set of content units by machine learning model(s) **120**.

(18) As further shown in FIG. 1, system **100** is implemented within a use environment including communication network **114** providing network communication links **116**, recommendation engine **122**, usage database **126**, user system **130** including display **132**, and user **118** of user system **130**. Also shown in FIG. 1 are multiple content units **136**, content compilation **128** composited by system **100** using software code **110** (hereinafter “composited content compilation **128**”), and usage data **134** corresponding to the engagement level of user **118** with composited content compilation **128**.

(19) Although the present application refers to one or more of software code **110**, machine learning model(s) **120**, and content and feature set database **112** as being stored in system memory **106** for conceptual clarity, more generally, system memory **106** may take the form of any computer-readable non-transitory storage medium. The expression “computer-readable non-transitory storage medium,” as defined in the present application, refers to any medium, excluding a carrier wave or other transitory signal that provides instructions to processing hardware **104** of computing platform **102**. Thus, a computer-readable non-transitory storage medium may correspond to various types of media, such as volatile media and non-volatile media, for example. Volatile media may include dynamic memory, such as dynamic random access memory (dynamic RAM), while non-volatile memory may include optical, magnetic, or electrostatic storage devices. Common forms of computer-readable non-transitory storage media include, for example, optical discs, RAM, programmable read-only memory (PROM), erasable PROM (EPROM), and FLASH memory.

(20) It is further noted that although FIG. 1 depicts software code **110**, machine learning model(s) **120**, and content and feature set database **112** as being co-located in system memory **106**, that representation is also merely provided as an aid to conceptual clarity. More generally, system **100** may include one or more computing platforms **102**, such as computer servers for example, which may be co-located, or may form an interactively linked but distributed system, such as a cloud-based system, for instance. As a result, processing hardware **104** and system memory **106** may correspond to distributed processor and memory resources within system **100**. Consequently, in some implementations, one or more of software code **110**, machine learning model(s) **120**, and content and feature set database **112** may be stored remotely from one another on the distributed memory resources of system **100**.

(21) Processing hardware **104** may include multiple hardware processing units, such as one or more central processing units, one or more graphics processing units, and one or more tensor processing units. By way of definition, as used in the present application, the terms “central processing unit” (CPU), “graphics processing unit” (GPU), and “tensor processing unit” (TPU) have their customary meaning in the art. That is to say, a CPU includes an Arithmetic Logic Unit (ALU) for carrying out the arithmetic and logical operations of computing platform **102**, as well as a Control Unit (CU) for retrieving programs, such as software code **110**, from system memory **106**, while a GPU may be implemented to reduce the processing overhead of the CPU by performing computationally intensive graphics or other processing tasks. A TPU is an application-specific integrated circuit (ASIC) configured specifically for artificial intelligence (AI) processes such as machine learning.

(22) In some implementations, computing platform **102** may correspond to one or more web servers, accessible over a packet-switched network such as the Internet, for example. Alternatively, computing platform **102** may correspond to one or more computer servers supporting a private wide area network (WAN), local area network (LAN), or included in another type of limited distribution or private network. Moreover, in some implementations, communication network **114** may be a high-speed network suitable for high performance computing (HPC), for example a 10 GigE network or an Infiniband network.

(23) Although user system **130** is shown as a desktop computer in FIG. **1**, that representation is provided merely as an example as well. More generally, user system **130** may be any suitable mobile or stationary computing device or system that includes display **132** and implements data processing capabilities sufficient to implement the functionality ascribed to user system **130** herein. For example, in other implementations, user system **130** may take the form of a laptop computer, tablet computer, or smartphone, for example.

(24) With respect to display **132** of user system **130**, display **132** may be implemented as a liquid crystal display (LCD), light-emitting diode (LED) display, organic light-emitting diode (OLED) display, quantum dot (QD) display, or any other suitable display screen that perform a physical transformation of signals to light. Furthermore, display **132** may be physically integrated with user system **130** or may be communicatively coupled to but physically separate from user system **130**. For example, where user system **130** is implemented as a smartphone, laptop computer, or tablet computer, display **132** will typically be integrated with user system **130**. By contrast, where user system **130** is implemented as a desktop computer, display **132** may take the form of a monitor separate from user system **130** in the form of a computer tower.

(25) The functionality of software code **110** will be further described by reference to FIG. **2**, which shows flowchart **200** presenting an exemplary method for use by system **100** to perform automated compositing of content compilations, according to one implementation. With respect to the actions outlined in FIG. **2**, it is noted that certain details and features have been left out of flowchart **200** in order not to obscure the discussion of the inventive features in the present application.

(26) Referring to FIG. **2** in combination with FIG. **1**, flowchart **200** begins with receiving multiple content units **136**, each one of the content units including a start descriptor for an initial content segment and an end descriptor for a last content segment (action **210**). Multiple content units **136** may span a wide range of themes and topics including wildlife and nature, fantasy, action-adventure, animation, live action, to name a few examples, as well as content having a host of different visual styles and audio tracks created by different artists in different settings and for different purposes. Moreover, multiple content units **136** may include clips of audio-video (AV) content, audio clips without video, or video clips without audio, for example. In addition, each content unit may include semantic features, such as a written synopsis, as well as metadata tags identifying keywords describing the included content. For instance, in the case of AV content including clips of episodic television (TV) or movie content, that metadata may identify genre type (action, adventure, comedy, fantasy, etc.), lead character type (pirate, princess, superhero, teacher,

criminal, etc.), or subject type (educational, coming of age, sci-fi, sports, racing etc.) of the content clips, to name a few examples.

(27) In some implementations, multiple content units **136** may be received by system **100** from user system **130** via communication network **114** and network communication links **116**, based on selection inputs to user system **130** by user **118**. For example, multiple content units **136** may be content units affirmatively selected by user **118** from a menu of predetermined curated content units that have been annotated to include the synopsis and keyword metadata tags described above. Alternatively, in some implementations multiple content units **136** may be received by system **100** from recommendation engine **122**, based on content consumption preferences of user **118** that are known or inferred by recommendation engine **122**. In those various implementations, multiple content units **136** may be received by software code **110**, executed by processing hardware **104** of computing platform **102**.

(28) With respect to the start descriptor and end descriptor of each of multiple content units **136**, it is noted that those start and end descriptors are generated from a feature set previously extracted from each content unit and stored in content and feature set database **112**. Referring to FIGS. **3A** and **3B**, FIG. **3A** shows diagram **300A** depicting extraction of an exemplary feature set from content unit **336** in the form of an exemplary AV content clip, according to one implementation, while FIG. **3B** shows diagram **300B** depicting generation of exemplary start descriptor **350a** and end descriptor **350b** of content unit **336**, according to one implementation. It is noted that content unit **336** corresponds in general to any or all of multiple content units **136**, and those corresponding features may share any of the characteristics attributed to content unit **336** by the present disclosure.

(29) As shown in FIG. **3A**, a variety of different analytical processes can be used to extract different features included in content unit **336**. For example, spectrogram analysis **338a** can be used to extract audio features **342** from content unit **336**, while one or more machine learning model based image extraction techniques **338b**, using one or more of machine learning model(s) **120** in FIG. **1** for example, may be used to extract image features **344**. In addition, various natural language processing based (NLP-based) techniques **338c** may be used to extract semantic features **346** and keyword features **348** from content unit **336**.

(30) The analytical processes performed on content unit **336** result in generation of a multi-valued floating point vector representation (i.e., feature vectors) for each of audio features **342**, image features **344**, semantic features **346**, and keyword features **348**. By way of example, and as noted above, signal processing techniques like spectrogram analysis **338a** of the audio track included in content unit **336** may be used to identify audio features **342**. A trained NN, such as a convolutional NN (CNN) may be used to identify image features **344**. For example, VGG-19 is a trained CNN that is very effective at object recognition. The initial layers of this CNN may be used to generate efficient and compact image feature representations for individual video frames.

(31) As described above, content unit **336** may include a written synopsis and metadata tags identifying keywords describing content unit **336**. Those text elements can be analyzed with NLP-based techniques **338c** such as Global Vectors for Word Representation (GloVe) and Bidirectional Encoder Representations from Transformers (BERT) word embeddings, WordNet, and term frequency-inverse document frequency (TF-IDF) feature generation. These NLP-based techniques enable representation of the clip description and its metadata in a feature vector space where they can be compared, clustered, and classified very efficiently. In addition, the word embedding techniques used allow for handling various categories of ontologies and can handle nuances when synonyms, hyponyms, hypernyms and even contronyms are present.

(32) Rather than utilizing a single set of feature vectors describing the entirety of content unit **336**, the present approach contemplates using two sets of feature vectors for each content unit **336**, a start feature vector for an initial content segment of content unit **336** and an end feature vector for a last content segment of the content unit. In some implementations, start and end feature vectors are

computed only for the audio and image features of content unit **336**. In those implementations, one feature vector each is computed for the semantic and keyword features using all of the available textual features of content unit **336**.

(33) As a specific example in use cases involving AV content units, video frames from a predetermined time duration or frame count from the beginning and end of content unit **336**, such as fifteen percent (15%) of the total time duration or frame count of content unit **336** for example, may be used to compute the start and end image feature vectors. For the audio feature vectors, another predetermined subset of the front half and the end half of the audio track of content unit **336** may be used, such as 50% of each of the front half and back half of the audio track, for example. It is noted that the exemplary percentages described above are hyperparameters that can be set and selectably modified by an administrator of system **100**.

(34) Once the feature vector sets for content unit **336** have been identified, a start descriptor for content unit **336** may be generated by concatenating or joining two or more of the start audio feature vector, the start image feature vector, the semantic feature vector for the entire content unit, and the keyword feature vector for the entire content unit together, end-to-end. Analogously, an end descriptor for content unit **336** may be generated by using one or both of the end audio and image feature vectors in similar combination with one or both of the same semantic and keyword feature vectors used to generate the start descriptor. This process is depicted in FIG. **3B**, which shows start descriptor **350a** and end descriptor **350b** of content unit **336**.

(35) As shown in FIG. **3B**, start descriptor **350a** includes start audio feature vector **352a** in combination with start image feature vector **354a**, semantic feature vector **356**, and keyword feature vector **358**, while end descriptor **350b** includes end audio feature vector **352b** in combination with end image feature vector **354b**, semantic feature vector **356**, and keyword feature vector **358**. Thus, for each content unit **336**, start descriptor **350a** and end descriptor **350b** may take the form of respective vectors in a multi-dimensional feature space. Moreover, as noted above, in some implementations, the content units corresponding to content unit **336** may be AV clips from a variety of different genres. In some of those implementations, as shown in FIG. **3B**, each of start descriptor **350a** and end descriptor **350b** may include two or more of an audio feature vector component, an image feature vector component, a keyword feature vector component, and a semantic feature vector component based on a written synopsis of the AV clip.

(36) Flowchart **200** further includes identifying start descriptor **350a** and end descriptor **350b** for each one of content units **136/336** (action **220**). Action **220** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, by accessing content and feature set database **112**.

(37) Flowchart **200** further includes selecting a first content unit of multiple content units **136/336** for beginning a content compilation (action **230**). The selection of the first content unit in action **230** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**. In various implementations, the selection of the first content unit for beginning the content compilation may be performed randomly, or may be based on a first content identifier included in multiple content units **136/336**. That is to say, in some implementations the first content unit selected in action **230** may be selected from within a set of contents suggested by recommendation engine **122** or by user **118**. However, in other implementations, selection of the first content unit for beginning the content compilation may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, in the manner described below by reference to FIGS. **4A** and **4B**.

(38) FIG. **4A** shows a portion **400A** of exemplary multi-dimensional feature space **460** including feature space landmarks **464** represented in FIG. **4A** by exemplary feature space landmarks **464a**, **464b**, and **464c**, according to one implementation. Also shown in FIG. **4A** are clusters of content units mapped onto multi-dimensional feature space **460** and represented in FIG. **4A** by exemplary content clusters **462a** and **462b**. It is noted that the content units mapped into multi-dimensional

feature space **460**, in FIG. **4A**, correspond in general to content unit **336**. Consequently, the content units included in content the clusters represented by exemplary content clusters **462a** and **462b** may share any of the characteristics attributed to content unit **336** by the present disclosure, and vice versa. It is further noted that the dimensions of multi-dimensional feature space **460** may correspond respectively to some, or all, of the feature vectors characterizing content unit **336**. The multi-dimensional feature space may have 3 modes or modalities (audio, visual, and textual), for example with many dimensions in each modality.

(39) As part of the process of training machine learning model(s) **120**, in FIG. **1**, a corpus of expertly composed content compilations may be aggregated. The content compilations included in this corpus may be produced by expert human content editors, and are known to be well formed and suitable for use as “best-of-class” exemplars for the content compilations to be composited by system **100**. The corpus of expertly composed content compilations is aggregated so as to be sizeable and diverse enough to cover substantially all of the visual styles, audio themes, topics, and semantic features that content unit **336** would typically depict. There may also be diversity in how the content units are composited together to form a satisfying narrative arc embodying qualities of good storytelling. For example a good content compilation may have an intriguing start, suspenseful middle, and an exciting end.

(40) Referring to FIG. **4A**, the content units included in each expertly composited content compilation are plotted in multi-dimensional feature space **460**. A clustering algorithm may then be run that finds content clusters. e.g., content clusters **462a** and **462b**, as well as cluster centers. i.e., superclusters of the content clusters (hereinafter “feature space landmarks”) represented in FIG. **4A** by exemplary feature space landmarks **464**. It is noted that the number of content clusters and feature space landmarks to be identified by the clustering algorithm is a hyperparameter that can be set and selectably modified by an administrator of system **100**.

(41) Each expertly composited content compilation can be visualized as a trajectory among the feature space landmarks **464** of multi-dimensional feature space **460**. That is to say, in some implementations, each of content units **136/336** may further include a feature space landmark **464** identifier corresponding to one or more of multiple content clusters in multi-dimensional feature space **460**, and each expertly composited content compilation can be visualized as a trajectory along the sequence of feature space landmarks **464** identified by the content units composited into that content compilation. An example of such a trajectory, with a merely exemplary length of five content units in the interests of conceptual clarity, is shown by diagram **400B** in FIG. **4B**. FIG. **4B** shows content compilation trajectory **466** starting at feature space landmark **464a**, continuing sequentially to respective feature space landmarks **464b**, **464c**, and **464d**, and ending at feature space landmark **464e**.

(42) In some implementations, selection of the first content unit in action **230** may be based on the first content units in the expertly composited content compilations serving as exemplars. That is to say, action **230** may include determining a first desirable feature space landmark for the first content unit, and selecting the first content unit based on the first desirable feature space landmark.

(43) Merely by way of example, let it be assumed that each expertly composited content compilation includes a first content unit having a feature space landmark identifier for one of feature space landmarks **464a**, **464b**, or **464c**, but no other feature space landmarks. Let it be further assumed that feature space landmark **464a** is identified most often chosen by the first content element in an expertly composited content compilation, followed by feature space landmark **464b**, which in turn is identified more frequently than feature space landmark **464c**.

(44) To select the first content unit in action **230**, the feature space landmark identifier of each of content units **136/336** may be obtained, and the weighted distance of each feature space landmark identified by those content units from each of the expert selected starting feature space landmarks **464a**, **464b**, and **464c** may be determined. It is noted that the weights applied to the distances described above may be the inverse ratios of the number of times each of feature space landmarks

464a, **464b**, and **464c** was identified by the first content unit in the expertly composited content compilations. Finally, the weighted distances of each of the feature space landmarks identified by the content units from each of feature space landmarks **464a**, **464b**, and **464c** can be averaged. In this particular use case, the characteristic determining the desirability of the feature space landmark of the content unit to be selected as the first content unit for beginning the content compilation in action **230** may be its averaged weighted distance from feature space landmarks **464a**, **464b**, and **464c**. The content unit that identifies the feature space landmark having that shortest averaged weighted distance from feature space landmarks **464a**, **464b**, and **464c** may then be selected as the first content unit in action **230**.

(45) Flowchart **200** further includes determining similarity metrics each of which compares end descriptor **350b** of the first content unit selected in action **230** with start descriptor **350a** of each of the other content units of multiple content units **136/336** (action **240**). Action **240** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**. It is noted that although the exemplary implementations described below refer to the similarity metrics determined in action **240** as corresponding to Euclidean distances, that characterization is merely provided by way of example. In other implementations, the similarity metrics determined in action **240** may include Manhattan distances (also known as L1-distances), or may include other similarity measures, such as cosine similarity for example. In various implementations, these similarity metrics determined in action **240** measure the distance between vectors in a multi-dimensional vector space as a means of quantifying their similarity.

(46) Referring to FIG. 5A. FIG. 5A shows diagram **501A** depicting compositing of content compilation **556**, according to one exemplary implementation. Content compilation **556** is shown to be partially composited and to include first content unit **536a**, second content unit **536b**, and third content unit **536c**. Also shown in FIG. 5A are other, not yet composited content units **536d**, **536e**, **536f**, and **536g** (hereinafter “content units **536d-536g**”), as well as distances **538d**, **538e**, **538f**, and **538g** (hereinafter “distances **538d-538g**”) of a start descriptor of each of respective content units **536d-536g** from the end descriptor of third content unit **536c**.

(47) It is noted that, according to the exemplary implementation shown in FIG. 5A, distances **538d-538g** correspond to the similarity metrics determined in action **240**. Thus, referring to FIGS. 2, 3B, and 5A in combination, action **240** may include determining distances **538d-538g** of end descriptor **350b** of third content unit **356c** from start descriptor **350a** of each of respective content units **536d-536g**. Distances **538d-538g** may be the Euclidean distances of the start descriptor **350a** of each of respective content units **536d-536g** from the end descriptor **350b** of third content unit **536c** in multi-dimensional feature space **460** described above. It is further noted that each of first content unit **536a**, second content unit **536b**, third content unit **536c**, and content units **536d-536g** correspond in general to content unit **336**, in FIG. 3A. Thus, each of first content unit **536a**, second content unit **536b**, third content unit **536c**, and content units **536d-536g** may share any of the characteristics attributed to content unit **336** by the present disclosure, and vice versa.

(48) Flowchart **200** further includes ranking, using the similarity metrics determined in action **240**, content units **536d-536g** with respect to one another (action **250**). For example, where the similarity metrics determined in action **240** are distances **538d-538g**, the ranking performed in action **250** may rank content units **536d-536g** based on respective distances **538d-538g**, with content unit **536d** associated with least distance **538d** being ranked highest. i.e., first, and content unit **536g** associated with greatest distance **538g** being ranked lowest, i.e., last. The ranking of content units **536d-536g** in action **250** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**.

(49) Flowchart **200** further includes selecting, based on the ranking performed in action **250**, a second content unit of multiple content units **136/336** (action **260**). Action **260** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, and in some implementations may be performed using machine learning model(s) **120**. The specific

process depicted by diagram **501A** is the selection of a fourth content unit for continuation of the compositing of content compilation **556** from among content units **536d-536g**. According to the exemplary implementation shown in FIG. 5A, the content unit among content units **536d-536g** having the shortest distance among distances **538d-538g**, i.e., content unit **536d**, is selected as the fourth content unit for continuing compositing of content compilation **556**.

(50) However, it is noted that a process analogous to that described by reference to diagram **501A** may be used to select second content unit **536b** based on the distance between the end descriptor of first content unit **536a** and the start descriptor of second content unit **536b**, as well as to select third content unit **536c** based on the distance between the end descriptor of second content unit **536b** and the start descriptor of third content unit **536c**.

(51) As seen in the approach described above by reference to FIG. 5A, at each step in the process software code **110**, executed by processing hardware **104**, must make a heuristic choice to select the next content unit of content compilation **556**. In some implementations, software code **110** may utilize machine learning model(s) to produce content compilation **556** and to learn over time to improve its own selection performance. In some such implementations, machine learning model(s) **120** receives as input the end descriptor of the last content unit already in partially composited content compilation **556**, i.e., third content unit **536c**, and all of the start descriptors of a predetermined number “k” of ranked content units **536d-536g** joined end-to-end as one complete input feature vector. It is noted that the integer value k is a hyperparameter for the system that can be set and selectably modified by an administrator of system **100**. It is further noted that top k candidate content units are selected on the same distance based approach described above. Thus, the top k content units are the k closest clips to the last clip already composited into content compilation **556** in feature vector space **460**.

(52) Machine learning model(s) **120** produces k outputs, representing the 1-of-k choices that must be made in order to select the next content unit for continuing content compilation **556**. The learning objective for machine learning model(s) **120** is to pick a best next content unit from among the best k candidates available to it.

(53) As noted above by reference to FIGS. 4A and 4B, in some implementations, each of first content unit **536a**, second content unit **536b**, third content unit **536c**, and content units **536d-536g** further includes a feature space landmark identifier corresponding to one or more of the content clusters represented by content clusters **462a** and **462b** in multi-dimensional feature space **460**. In those implementations, processing hardware **104** may further execute the software code **110** to identify the feature space landmark for each of first content unit **536a**, second content unit **536b**, third content unit **536c**, and content units **536d-536g**, and to select sequential content units for compositing of content compilation **556** further based on those feature spaced landmarks. That is to say, processing hardware **104** may further execute the software code **110** to determine, based on the first feature space landmark for first content unit **536a**, a desirable second feature space landmark for second content unit **536b**, and to select second content unit **536b** further based on that desirable second feature space landmark in action **260**. Referring to FIG. 5B, FIG. 5B depicts compositing of content compilation **556** using such an approach.

(54) FIG. 5B shows diagram **501B** depicting compositing of content compilation **556**, according to another exemplary implementation. Content compilation **556** is shown to be partly composited and to include first content unit **536a**, second content unit **536b**, and third content unit **536c**. Also shown in FIG. 5B are other, not yet composited content units **536d**, **536e**, **536f**, **536g**, and **536h** (hereinafter “content units **536d-536h**”), as well as distances **538d**, **538e**, **538f**, **538g**, and **538h** (hereinafter “distances **538d-538h**”) of a start descriptor of each of respective content units **536d-536h** from the end descriptor of third content unit **536c**. In addition, diagram **501B** shows trajectory **566** of content compilation **556**. Trajectory **566** begins at feature space landmark **564a** identified by first content unit **536a**, continues through feature space landmarks **564b** and **564c** identified by respective second and third content units **536b** and **536c**, and continues through feature space

landmarks **564d**, **564e**, **564f**, and **564g**, respectively, before ending at feature space landmark **564h**.
(55) Each of feature space landmarks **564a**, **564b**, **564c**, **564d**, **564e**, **564f**, **564g**, and **564h** (hereinafter “feature space landmarks **564a-564h** corresponds in general to any of feature space landmarks **464a**, **464b**, **464c**, **464d**, and **464e** (hereinafter “feature space landmarks **464a-464e**”) shown in FIG. 4A or 4B. Thus, each of feature space landmarks **564a-564h** may share any of the characteristics attributed to any of feature space landmarks **464a-464e** by the present disclosure, and vice versa.

(56) Referring to FIGS. 1 and 5B in combination, in some implementations, machine learning model(s) **120** have been trained as sequence prediction model(s) using the corpus of expertly composited content compilations described above by reference to FIGS. 4A and 4B. When implemented as sequence prediction model(s), machine learning model(s) may take the form of one or more of a recurrent neural network (RNN), having a long short-term memory (LSTM) architecture, for example, or as any finite-state transducer. Machine learning model(s) are configured to predict the next feature space landmark in trajectory **566** of content compilation **556**, given the previous partial order of feature space landmarks.

(57) The prediction takes the form of probabilities, one each for each of the possible feature space landmarks. The feature space landmark with the highest probability is identified as the next feature space landmark in the sequence of trajectory **566**. Consider the last content unit composited into content compilation **556** thus far, i.e., third content unit **536c**. In order to select the next clip, i.e., the fourth content unit for content compilation **556**, feature space landmark **564c** identified by third content unit **536c**, as well as, in some implementations, one or both of feature space landmarks **564a** and **564b** identified by respective first content unit **536a** and second content unit **536b**, can be analyzed by machine learning model **120** to predict a list of desirable feature space landmarks for use as next feature space landmark **564d** of trajectory **566**. From that list, the feature space landmark having the highest probability may be selected as feature space landmark **564d**. Then, from the ranking of the content units described above by reference to action **250** and FIG. 5A, based on distances **538a-538h**, the highest ranked content unit including the feature space landmark identifier of the feature space landmark selected as feature space landmark **564d** may then be selected as the next content unit. i.e., the fourth content unit.

(58) As shown in FIG. 5B, the feature space landmark identifier of feature space landmark **564d** selected as the next feature space landmark of trajectory **566** is Lmk-B. Of ranked content units **536d-536h**, only content unit **536g** includes feature space landmark identifier Lmk-B. Consequently, despite being ranked lower than content units **536d**, **536e**, and **536f**, content unit **536g** may be selected as the fourth content unit of content compilation **556**.

(59) It is possible that none of content units **536d-536h** includes the feature space landmark identifier of the feature space landmark having the highest probability predicted by machine learning model(s) **120**. Under those circumstances, the feature space landmark having the second highest probability may be selected as feature space landmark **564d** of trajectory **566**, and the fourth content unit may then be selected based on being the highest ranked of content units **536d-536e** including the feature space landmark identifier for the second highest probability feature space landmark may be selected as the fourth content unit of content compilation **556**. This selection process may continue until the fourth content unit of content compilation is successfully identified.

(60) It is noted that although the specific example shown and described by reference to FIG. 5B refers to selection of the fourth content unit of content compilation **556**, a process analogous to that described by reference to diagram **501B** may be used to select second content unit **536b**, third content unit **536c**, and so forth. Thus, in some implementations, action **260** may include determining, based on the first feature space landmark identified by first content unit **536a**, desirable second feature space landmark **564b** for second content unit **536b**, and selecting second content unit **536b** further based on desirable second feature space landmark **564b**. Moreover, in

some implementations, second content unit **536b** may be selected by software code **110**, executed by processing hardware **104** of computing platform **102**, and using machine learning model(s) **120**. (61) In some implementations, flowchart **200** can continue and conclude with compositing of content compilation **556** using second content unit **536b** (action **270**). As noted above, the automated method outlined by flowchart **200** follows a heuristic approach to composite content compilation **556** beginning with first content unit **536a** selected in action **230**. The process of compositing of content compilation **556** can further include selecting third content unit **536c**, selecting content unit **536d** as the fourth content unit of content compilation **556**, selecting one of content units **536e**, **536f**, or **536g** as the fifth content unit of content compilation **556**, and so forth. Action **270** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, and results in production of composited content compilation **128**.

(62) Although in some implementations flowchart **200** can conclude with action **270**, in implementations in which second content unit **536b** is selected using machine learning model(s) **120**, flowchart **200** may further include obtaining usage data **134** corresponding to an engagement level of user **118** with composited content compilation **128** (action **280**), and training machine learning model(s) **120** using usage data **134**, resulting in an improved selection performance by machine learning model(s) **120** (action **290**). As noted above, usage data **134** corresponds to the engagement level of user **118** with composited content compilation **128**. Such an engagement level may be determined based on how much of composited content compilation **128** user **118** consumes: either user **118** consumes composited content compilation **128** in its entirety, which can be taken as a positive sign, or if user **118** abandons it part way through, that could be construed as some measure of dislike or non-engagement.

(63) FIG. **6A** shows diagram **601A** depicting production of an implicit signal for improving the selection performance of the system shown in FIG. **1**, based on usage data, while FIG. **6B** shows diagram **601B** depicting production of an implicit signal for improving content sequencing performance for compositing of a content compilation, based on usage data. As shown in FIG. **6A**, content compilation **656** is shown to be partly composited and to include first content unit **636a**, second content unit **636b**, and third content unit **636c**. Also shown in FIG. **6A** are other, not yet composited content units **636d**, **636e**, **636f**, **636g**, and **636h** (hereinafter “content units **636d-636h**”), as well as distances **638d**, **638e**, **638f**, **638g**, and **638h** (hereinafter “distances **638d-638h**”) of a start descriptor of each of respective content units **636d-636h** from the end descriptor of third content unit **636c**. In addition, diagram **601A** shows exemplary reinforcement based machine learning model **620**.

(64) It is noted that each of first content unit **636a**, second content unit **636b**, third content unit **636c**, and content units **636d-636h** correspond in general to content unit **336**, in FIG. **3A**. Thus, each of first content unit **636a**, second content unit **636b**, third content unit **636c**, and content units **636d-636h** may share any of the characteristics attributed to content unit **336** by the present disclosure, and vice versa. It is further noted that distances **638d-638g**, in FIG. **6A**, correspond respectively in general to distances **538d-538g**, in FIGS. **5A** and **5B**, while distance **638h** corresponds in general to distance **538h** in FIG. **5B**. Moreover, reinforcement based machine learning model **620** corresponds in general to machine learning model(s) **120**, in FIG. **1**, and those corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure.

(65) FIG. **6B** shows content compilation **628** composited by system **600** for automated compositing of content compilations, and usage data obtained by system **600**. Also shown in FIG. **6B** is corpus **615** of content clips available for compositing. System **600**, content compilation **628**, and usage data **634** correspond respectively in general to system **100**, content compilation **128**, and usage data **134**, in FIG. **1**. Thus, system **600**, content compilation **628**, and usage data **634** may share any of the characteristics attributed to system **100**, content compilation **128**, and usage data **134** by the present disclosure, and vice versa.

(66) Using a predetermined threshold for the engagement level, usage data **134/634** may be used to produce an implicit signal that system **100/600** can use to improve the sequencing performance of reinforcement based machine learning model **620**. As discussed above, at each step in the process of compositing content compilation **128/628**, system **100/600** must make a heuristic choice to select the next clip in the sequence. According to the exemplary implementation shown in FIG. **6A** reinforcement based machine learning model **620** can, over the course of time, learn the best clip to insert at the present position that works over and above that of the basic heuristic process.

(67) For example, reinforcement based machine learning model **620** model may receive as input the end descriptor of the last clip already in partially constructed sequence so far, e.g., third content clip **636c**, and all of the start descriptors of the top k candidate clips (e.g., content clips **636d-636h**), joined end-to-end as one complete input feature vector. Here, the number k is a hyperparameter for system **100/600** that can be set and selectably modified by an administrator of system **100/600**. It is noted that the top k candidate clips may be selected on the same Euclidean or other distance based approach outlined above by examining all content clips that are not already include in partially composited content compilation **656**. This means that the top k content clips are the k closest content clips to the last clip already in the sequence (e.g., third content clip **636c**), in feature vector space. The input feature vector applied to reinforcement based machine learning model **620** model represents the ‘current state of the world’ in reinforcement based machine learning parlance. Reinforcement based machine learning model **620** has k outputs, representing the 1-of-k choices that must be made in selecting the next content clip in the sequence. The learning objective for reinforcement based machine learning model **620** is to select a next best content clip from the k candidates available to it.

(68) After the process depicted in FIG. **6A** has constructed the full sequence using a combination of heuristics for short-listing the top-k candidate content clips (e.g. as referenced in FIG. **5A**) and a trainable landmark traversal model to select a best next content clip from among those candidates at each step in the sequence (e.g. as referenced in FIG. **5B**), the final composited content compilation **128/628** is presented to user **118**. Based on usage data **134/634** describing how user **118** engages with the composition, for example the percentage of content compilation **128/628** viewed by user **118** before user **118** abandons content compilation **128/628**, an implicit signal can be generated and can be used for further training or retraining of reinforcement based machine learning model **620**. For example, as depicted in FIG. **6B**, a reward or penalty may be assessed for composited content compilation **128/628**, characterized by real numbers for instance, e.g., +1.0 for reward. -1.0 as a penalty. This reward or penalty can be applied or back-propagated through a reinforcement learning model of machine learning model **620** so that reinforcement based machine learning model **620** can, over the course of multiple iterations, learn to composite content compilations that are sequenced such that user **118** is likely to consume them in their entirety and possibly return to again and again.

(69) As shown in FIG. **1**, usage data **134/634** may be obtained in action **280** from usage database **126** or user system **130**, via communication network **114** and network communication links **116**. Actions **280** and **290** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**. One advantage of the approach described by reference to FIGS. **6A** and **6B** is that even an initially untrained machine learning model can make effective choices from the candidate clips, because the candidate clips are already short-listed based on desirable characteristics. e.g., closeness to the last clip already in the sequence. Over the course of time as the machine learning model receives feedback based on usage data **134/634**, the machine learning model can continue to improve.

(70) Referring to FIG. **7**, FIG. **7** shows flowchart **700** presenting an exemplary method for use by system **100** to perform automated compositing of content compilations, according to another implementation. With respect to the actions outlined in FIG. **7**, it is noted that certain details and features have been left out of flowchart **700** in order not to obscure the discussion of the inventive

features in the present application.

(71) Referring to FIGS. 1, 3A, and 3B in conjunction with FIG. 7, flowchart **700** begins with receiving multiple content units **136/336**, each one of the content units **136/336** including start descriptor **350a** for an initial content segment and end descriptor **350b** for a last content segment (action **710**). Action **710** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, in the manner described above by reference to action **210** of flowchart **200**.

(72) Flowchart **700** further includes identifying start descriptor **350a** and end descriptor **350b** for each one of content units **136/336** (action **720**). Action **720** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, by accessing content and feature set database **112**.

(73) Flowchart **700** further includes identifying multiple candidate content compilations, each of the candidate compilations including multiple content units **136/336** in a different temporal sequence (action **730**). For example, where multiple content units **136/336** include three content units: e.g., content unit A, content unit B, and content unit C, action **730** may result in identification of three factorial ($3!$), i.e., six, candidate compilations: ABC, ACB, BAC, CAB, BCA, CBA, while any integer number “ n ” of multiple content units **136/336** may result in identification of up to $n!$ content compilations. Thus, the candidate compilations identified in action **730** may include all $n!$ permutations of multiple content units **136/336**, or any subset thereof. Moreover, the different permutations of candidate compositions can be used to compile a set of similarity metrics with which each candidate composition can be ranked and the best one be identified, as described below by reference to actions **740** and **750**. It is noted that such similarity metrics may be compiled using one or both of the baseline heuristic approach described above by reference to FIG. 5A, or the landmark traversal approach described above by reference to FIG. 5B. Action **730** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**.

(74) Flowchart **700** further includes, for each one of the candidate content compilations identified in action **730**, determining, by comparing the end descriptor of each of the content units to the start descriptor of the next content unit in the temporal sequence, a score or a continuity score for that candidate content compilation, resulting in multiple continuity scores (action **740**). In one implementation in which each candidate content compilation includes “ n ” content units, for example, the distance in a multi-dimensional feature space separating the end descriptor of each of the first 1 to $(n-1)$ content units to the start descriptor of the next content unit in the temporal sequence of each candidate content compilation may be computed, in a manner analogous to that described above by reference to FIG. 5A. In that implementation, the continuity score may be determined to be the sum of those distances for each candidate content compilation.

(75) As noted above by reference to FIGS. 4A, 4B, and 5B, in some implementations, each of multiple content units **136/336** further includes a feature space landmark identifier corresponding to one or more content clusters in a multi-dimensional feature space. In those implementations, the continuity score for each of the candidate content compilations may further be determined by comparing the feature space landmark identifier of each content unit of the candidate content compilation to the feature space landmark of the next content unit in the temporal sequence of that candidate content compilation. For example, based on the comparison of the feature space landmark identifiers, a trajectory of each candidate content compilation in the multi-dimensional feature space may be determined, and that trajectory could be compared to the trajectories of the corpus of expertly composited content compilations described above. Action **640** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, and in some implementations, using machine learning model(s) **120**.

(76) In some implementations, flowchart **700** may continue and conclude with selecting, based on the continuity scores determined in action **740**, a best content compilation from among the candidate content compilations (action **750**). It is noted that, depending on the specific way in

which the continuity score is determined, the best candidate content compilation may be the candidate content compilation having the highest, or the lowest, continuity score. For example, in implementations in which the sum of the distances separating the end descriptor of each of the first 1 to (n-1) content units to the start descriptor of the next content unit in the temporal sequence of each candidate content compilation determines the continuity score, the candidate content compilation having the lowest score. i.e., smallest sum of distances, may be selected as the best content compilation. Action **750** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**.

(77) Although in some implementations flowchart **700** can conclude with action **750**, in implementations in which the best content compilation is selected in action **750** using machine learning model(s) **120**, flowchart **700** may further include obtaining usage data **134** corresponding to an engagement level of user **118** with the selected best content compilation (action **760**), and training machine learning model(s) **120** using usage data **134**, resulting in an improved selection performance by machine learning model(s) **120** (action **770**). Actions **760** and **770** may be performed by software code **110**, executed by processing hardware **104** of computing platform **102**, in a manner analogous to actions **280** and **290** described above.

(78) With respect to the methods outlined by flowcharts **200** and **700**, it is noted that actions **210**, **220**, **230**, **240**, **250**, **260**, and **270**, or actions **210**, **220**, **230**, **240**, **250**, **260**, **270**, **280**, and **290**, or actions **710**, **720**, **730**, **740**, and **750**, or actions **710**, **720**, **730**, **740**, **750**, **760**, and **770**, may be performed in an automated process from which human involvement can be omitted.

(79) Thus, the present application discloses automated systems and methods for compositing content compilations that address and overcome the deficiencies in the conventional art. As described above, according to the present automated compositing solution, information about features of content units, such as their audio features, the images they include, and their semantic features, for example, are extracted and used to automate the compositing of the content units in a manner that produces results comparable in quality to those produced by an expert human content editor. In contrast to conventional techniques for creating content compilations, the present solution can advantageously work with content that spans a wide range of themes and topics, as well as a host of visual styles and audio tracks created by different artists in different settings and for different purposes. In addition, according to the present compositing solution a user supplied template or layout, which is typically relied upon in conventional content compilation techniques, is neither sought nor utilized. As a result, the present solution advantageously enables the automated production of a coherent content compilation without abrupt switches in visual style, story arc, or audio effects.

(80) From the above description it is manifest that various techniques can be used for implementing the concepts described in the present application without departing from the scope of those concepts. Moreover, while the concepts have been described with specific reference to certain implementations, a person of ordinary skill in the art would recognize that changes can be made in form and detail without departing from the scope of those concepts. As such, the described implementations are to be considered in all respects as illustrative and not restrictive. It should also be understood that the present application is not limited to the particular implementations described herein, but many rearrangements, modifications, and substitutions are possible without departing from the scope of the present disclosure.

Claims

1. A system comprising: a computing platform including a processing hardware and a system memory storing a software code; the processing hardware configured to execute the software code to: receive a plurality of content units, the plurality of content units comprising at least one of: (i) content having a plurality of different themes, or (ii) content on a plurality of different topics, each

one of the plurality of content units including a start descriptor for an initial content segment and an end descriptor for a last content segment; identify the start descriptor and the end descriptor for each one of the plurality of content units; select a first content unit of the plurality of content units for beginning a content compilation; determine a plurality of similarity metrics each comparing the end descriptor of the first content unit with the start descriptor of a respective one of other content units of the plurality of content units; rank, using the plurality of similarity metrics, the one other content unit of the plurality of content units with respect to one another; select, based on the rank, a second content unit of the plurality of content units; and composite the content compilation using the second content unit.

2. The system of claim 1, wherein each one of the plurality of content units further includes a feature space landmark identifier corresponding to one or more of a plurality of content clusters in a multi-dimensional feature space, and wherein the processing hardware is further configured to execute the software code to: identify the feature space landmark for each of the plurality of content units; and determine a first desirable feature space landmark for the first content unit; wherein the first content unit is selected based on the first desirable feature space landmark.

3. The system of claim 1, wherein each one of the plurality of content units further includes a feature space landmark identifier corresponding to one or more of a plurality of content clusters in a multi-dimensional feature space, and wherein the processing hardware is further configured to execute the software code to: identify the feature space landmark for each of the plurality of content units; and determine, based on a first feature space landmark for the first content unit, a desirable second feature space landmark for the second content unit; wherein the second content unit is selected further based on the desirable second feature space landmark.

4. The system of claim 3, wherein the second content unit is selected using a machine learning model, and wherein the processing hardware is further configured to execute the software code to: obtain usage data corresponding to an engagement level of a user with the composited content compilation; and train, using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.

5. The system of claim 1, wherein the second content unit is selected using a machine learning model, and wherein the processing hardware is further configured to execute the software code to: obtain usage data corresponding to an engagement level of a user with the composited content compilation; and train, using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.

6. The system of claim 1, wherein, for each of the plurality of content units, the start descriptor and the end descriptor comprise respective vectors in a multi-dimensional feature space, and wherein the plurality of similarity metrics each corresponds to a distance between the end descriptor for the first content unit and the start descriptor for the respective one of other content units of the plurality of content units in the multi-dimensional vector space.

7. The system of claim 6, wherein the plurality of content units comprise audio-video (AV) clips of content from a plurality of different genres, and wherein for each of the plurality of AV clips, each of the start and end descriptors comprises an audio feature vector component, an image feature vector component, and a keyword feature vector component.

8. The system of claim 7, wherein for each of the plurality of AV clips, each of the start and end descriptors further comprises a semantic feature vector component based on a written synopsis of the AV clip.

9. A method for use by a system including a computing platform having a processing hardware, and a system memory storing a software code, the method comprising: receiving, by the software code executed by the processing hardware, a plurality of content units, the plurality of content units comprising at least one of: (i) content having a plurality of different themes, or (ii) content on a plurality of different topics, each one of the plurality of content units including a start descriptor for an initial content segment and an end descriptor for a last content segment; identifying, by the

software code executed by the processing hardware, the start descriptor and the end descriptor for each one of the plurality of content units; selecting, by the software code executed by the processing hardware, a first content unit of the plurality of content units for beginning a content compilation; determining, by the software code executed by the processing hardware, a plurality of similarity metrics each comparing the end descriptor of the first content unit with the start descriptor of a respective one of other content units of the plurality of content units; ranking, by the software code executed by the processing hardware and using the plurality of similarity metrics, the one other content unit of the plurality of content units with respect to one another; selecting, by the software code executed by the processing hardware and based on the ranking, a second content unit of the plurality of content units; and compositing, by the software code executed by the processing hardware, the content compilation using the second content unit.

10. The method of claim 9, wherein each one of the plurality of content units further includes a feature space landmark identifier corresponding to one or more of a plurality of content clusters in a multi-dimensional feature space, the method further comprising: identifying, by the software code executed by the processing hardware, the feature space landmark for each one of the plurality of content units; and determining, by the software code executed by the processing hardware, a first desirable feature space landmark for the first content unit; wherein the first content unit is selected based on the first desirable feature space landmark.

11. The method of claim 9, wherein each one of the plurality of content units further includes a feature space landmark identifier corresponding to one or more of a plurality of content clusters in a multi-dimensional feature space, the method further comprising: identifying, by the software code executed by the processing hardware, the feature space landmark for each one of the plurality of content units; and determining, by the software code executed by the processing hardware and based on a first feature space landmark for the first content unit, a desirable second feature space landmark for the second content unit; wherein the second content unit is selected further based on being the desirable second feature space landmark.

12. The method of claim 11, wherein the second content unit is selected using a machine learning model, the method further comprising: obtaining, by the software code executed by the processing hardware, usage data corresponding to an engagement level of a user with the composited content compilation; and training, by the software code executed by the processing hardware and using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.

13. The method of claim 9, wherein the second content unit is selected using a machine learning model, the method further comprising: obtaining, by the software code executed by the processing hardware, usage data corresponding to an engagement level of a user with the composited content compilation; and training, by the software code executed by the processing hardware and using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.

14. The method of claim 9, wherein, for each of the plurality of content units, the start descriptor and the end descriptor comprise respective vectors in a multi-dimensional feature space, and wherein the plurality of similarity metrics each corresponds to a distance between the end descriptor for the first content unit and the start descriptor for the respective one of other content units of the plurality of content units in the multi-dimensional vector space.

15. The method of claim 14, wherein the plurality of content units comprise audio-video (AV) clips of content from a plurality of different genres, and wherein for each of the plurality of AV clips, each of the start and end descriptors comprises an audio feature vector component, an image feature vector component, and a keyword feature vector component.

16. The method of claim 15, wherein for each of the plurality of AV clips, each of the start and end descriptors further comprises a semantic feature vector component based on a written synopsis of the AV clip.

17. A system comprising: a computing platform including a processing hardware and a system memory storing a software code; the processing hardware configured to execute the software code to: receive a plurality of content units, the plurality of content units comprising at least one of: (i) content having a plurality of different themes, or (ii) content on a plurality of different topics, each one of the plurality of content units including a start descriptor for an initial content segment and an end descriptor for a last content segment; identify the start descriptor and the end descriptor for each one of the plurality of content units; identify a plurality of candidate content compilations, each of the plurality of candidate compilations including the plurality of content units in a different temporal sequence; for each one of the plurality of candidate content compilations, determine, by comparing the end descriptor of each of the plurality of content units to the start descriptor of a next one of the plurality of content units in the different temporal sequence, a score for the one of the plurality of candidate content compilations, resulting in a plurality of scores; and select, based on the plurality of scores, a best content compilation from among the plurality of candidate content compilations.

18. The system of claim 17, wherein each one of the plurality of content units further includes a feature space landmark identifier corresponding to one or more of a plurality of content clusters in a multi-dimensional feature space, and wherein the score for the one of the plurality of candidate content compilations is further determined by comparing the feature space landmark identifier of each of the plurality of content units to the feature space landmark identifier of a next one of the plurality of content units in the different temporal sequence.

19. The system of claim 18, wherein the best content compilation is selected using a machine learning model, and wherein the processing hardware is further configured to execute the software code to: obtain usage data corresponding to an engagement level of a user with the best content compilation; and train, using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.

20. The system of claim 17, wherein the best content compilation is selected using a machine learning model, and wherein the processing hardware is further configured to execute the software code to: obtain usage data corresponding to an engagement level of a user with the best content compilation; and train, using the usage data, the machine learning model, resulting in an improved selection performance by the machine learning model.
